Practice Set 3 and Mock 10: Wrong-Answer Notes

CFA Level II · 12 wrong answers + 4 right answers you were not fully sure about, one per page

46 / 52
Practice set 3 (88.5%)

38 / 44
Mock 10 (86.4%)

Set 3 Q49-52 were marked against the version you sat (before the 11 Nov re-skin).

Mock 10 score by topic

Topic	Score
Alt	2/4
FSA	3/4
Ethics	3/4
Quant	3/4
Deriv	3/4
Econ	4/4
Equity	8/8
Corp	4/4
PM	4/4
FI	4/4

Questions at a glance

Q	Topic	You	Ans
Set 3 Q9	Ethics	C	B
Set 3 Q21	Alt	C	A
Set 3 Q34	PM	B	A
Set 3 Q39	Quant	C	B
Set 3 Q43	PM	C	A
Set 3 Q50	FI	A	B
Mock 10 Q1	FSA	C	B
Mock 10 Q9	Ethics	A	B
Mock 10 Q13	Quant	C	A
Mock 10 Q14	Quant	A	A
Mock 10 Q15	Quant	C	C
Mock 10 Q23	Deriv	B	B
Mock 10 Q24	Deriv	A	B
Mock 10 Q37	Alt	A	B
Mock 10 Q38	Alt	A	C
Mock 10 Q40	Alt	A	A

Set 3 Q9 **Standard V(B): separate fact from opinion, and say how uncertain a forecast is**

You **C** · Answer **B**

Ethics

VIGNETTE (RELEVANT PART, UNEDITED)

Ethics: Standards of Professional Conduct

Question 9. Scott works for a regional brokerage firm. He estimates that Walkton Industries will increase its dividend by US$1.50 a share during the next year. He realizes that this increase is contingent on pending legislation that would, if enacted, give Walkton a substantial tax break. The US representative for Walkton's home district has told Scott that, although she is lobbying hard for the bill and prospects for its passage are favorable, concern of the US Congress over the federal deficit could cause the tax bill to be voted down. Walkton Industries has not made any statements about a change in dividend policy. Scott writes in his research report, "We expect Walkton's stock price to rise by at least US$8.00 a share by the end of the year because the dividend will increase by US$1.50 a share. Investors buying the stock at the current time should expect to realize a total return of at least 15% on the stock." According to the Standards:

OPTIONS

A	Scott violated the Standards because he used material inside information.		Not inside information: the facts came from a public representative and the company said nothing, so there is no material nonpublic information.
B	Scott violated the Standards because he failed to separate opinion from fact.	✓ Correct	Correct: V(B). The report states the dividend rise and an $8 price gain as certain, although both depend on the bill passing.
C	Scott violated the Standards by basing his research on uncertain predictions of future government action.	✗ Your pick	Your pick: basing research on an uncertain prediction of government action is allowed. The violation is how it was communicated.

Trigger "contingent on pending legislation"
"could cause the tax bill to be voted down"
"because the dividend will increase by US$1.50"

Rule next time: A forecast that depends on an uncertain event must not be written as if it were certain. V(B): separate fact from opinion and disclose the uncertainty. Relying on a prediction is fine; presenting it without the conditions is the violation.

FORMULA / RULE

> V(B): distinguish fact from opinion; include the important factors and limits of the analysis
> Words that signal a V(B) problem: "will", "at least", "guaranteed" attached to an uncertain event

WHY THE ANSWER IS B

1. The dividend increase is conditional on legislation that may fail.
2. The report says the price WILL rise "because the dividend will increase": opinion presented as fact.
3. **B**.

Apply to other questions: Ask: does the report show the condition behind the forecast? If not, V(B). Use of public, uncertain information is not itself a violation.

Set 3 Q21 NAVPS: capitalise NOI, then add other assets and subtract debt

You **C** · Answer **A**

Alt

VIGNETTE (RELEVANT PART, UNEDITED)

Castell values Alpenhaus REIT. Next year's net operating income is CHF 32 million and the market capitalisation rate for similar properties is 6.4%. Alpenhaus also holds CHF 18 million of cash and other tangible assets, and has CHF 290 million of debt. It has 10 million shares.

Castell notes that Alpenhaus is highly leveraged and trades at a large discount to NAVPS, and that its debt matures next year.

Question 21. Alpenhaus's NAVPS is closest to:

OPTIONS

A	CHF 22.80	✓ Correct	Correct: (500 + 18 − 290) ÷ 10 = CHF 22.80.
B	CHF 51.80		CHF 51.80 = (500 + 18) ÷ 10 adds the cash but forgets to subtract the debt.
C	CHF 50.00	✗ Your pick	Your pick: CHF 50.00 = 500 ÷ 10 is the property value per share. Cash and debt were left out.

Trigger “market capitalisation rate for similar properties is 6.4%”
“CHF 18 million of cash and other tangible assets”
“has CHF 290 million of debt”

Rule next time: NAV is the equity value: property value (NOI ÷ cap rate) + cash and other assets − debt. Only then divide by the shares.

FORMULA / RULE

> Property value = NOI ÷ market cap rate (direct capitalisation, not depreciated cost)
> NAV = property value + other assets − liabilities; NAVPS = NAV ÷ shares

WHY THE ANSWER IS A

1. Property value = 32 ÷ 0.064 = 500.0.
2. NAV = 500 + 18 − 290 = 228.0.
3. NAVPS = 228 ÷ 10 = CHF 22.80: **A**.

Apply to other questions: Every item in the vignette (cash, debt) is a step. Value ≠ NAV until the balance sheet items are applied.

Set 3 Q34 **Real rates follow REAL growth and its volatility, not nominal growth**

You **B** · Answer **A**

PM

VIGNETTE (RELEVANT PART, UNEDITED)

Point 3: Country 1 has nominal GDP growth of 8.5% and inflation of 3.0%, with low volatility of growth. Country 2 has nominal GDP growth of 6.0% and inflation of 0.5%, with high volatility of growth.

Question 34. Based on Point 3, real interest rates are most likely higher in:

OPTIONS

A	Country 2, because real growth is the same but growth is more volatile	✓ Correct	Correct: real growth is 5.5% in both, and Country 2's growth is more volatile, so its real rate is higher.
B	Country 1, because nominal growth is higher	✗ Your pick	Your pick: Country 1 has higher NOMINAL growth, but nominal growth includes inflation and says nothing about the real rate.
C	neither country, because real growth is the same		Equal real growth does not mean equal real rates: volatility of growth also matters.

Trigger “nominal GDP growth of 8.5% and inflation of 3.0%”
“nominal GDP growth of 6.0% and inflation of 0.5%”
“low volatility” vs “high volatility of growth”

Rule next time: Convert to real growth first: nominal − inflation. If real growth is equal, the country with more volatile growth has the higher real interest rate.

FORMULA / RULE

Real growth ≈ nominal growth − inflation

Real rate ↑ with higher expected real growth and with higher volatility of real growth

ITRS ↓ (better off in the future) → real rates ↑

WHY THE ANSWER IS A

1. Country 1: 8.5 − 3.0 = 5.5%. Country 2: 6.0 − 0.5 = 5.5%.
2. Equal, so compare volatility: Country 2 is higher.
3. Real rates higher in Country 2: **A**.

Apply to other questions: Subtract inflation before comparing growth rates, then look at volatility.

Set 3 Q39 **AIC is for forecasting, BIC for goodness of fit; for both the LOWER value is better**

You **C** · Answer **B**

Quant

VIGNETTE (RELEVANT PART, UNEDITED)

Exhibit 1: ANOVA for the Three-Factor Model

Source	Degrees of freedom	Sum of squares
Regression	3	0.0770
Error	56	0.0480
Total	59	0.1250

Greco also compares two candidate models. Model A has AIC −310 and BIC −298. Model B has AIC −314 and BIC −296.

Question 39. Based on AIC and BIC, which model is preferred for forecasting and which for goodness of fit?

OPTIONS

A	Model A for forecasting; Model B for goodness of fit		Reversed: A for forecasting and B for goodness of fit mixes up which criterion is for which purpose.
B	Model B for forecasting; Model A for goodness of fit	✓ Correct	Correct: AIC picks Model B (−314 < −310) for forecasting; BIC picks Model A (−298 < −296) for goodness of fit.
C	Model B for both	✗ Your pick	Your pick: Model B wins on AIC only. On BIC, Model A is lower.

Trigger “Model A has AIC −310 and BIC −298. Model B has AIC −314 and BIC −296”

Rule next time: AIC → forecasting (prediction), BIC → goodness of fit. Lower is better for both, and −314 is lower than −310.

FORMULA / RULE

AIC (lighter penalty for extra variables): better for forecasting
BIC: heavier penalty for extra variables: better for goodness of fit
Choose the model with the LOWEST value (with negative numbers, the more negative one)

WHY THE ANSWER IS B

1. AIC: −314 (B) < −310 (A) → B is better for forecasting.
2. BIC: −298 (A) < −296 (B) → A is better for goodness of fit.
3. **B**.

Apply to other questions: Compare each criterion separately; do not pick one model for both just because it wins on the first.

Set 3 Q43 **Discount rate for rents: tenant credit risk is in the discount rate; faster rent growth is in the cash flow**

You **C** · Answer **A**

PM

VIGNETTE (RELEVANT PART, UNEDITED)

Lucia Ferrante makes three further observations.

Observation 3: An investor compares a building leased to a government agency with a similar building leased to commercial tenants.

Question 43. For Observation 3, the commercial building's rental income should most likely be discounted at:

OPTIONS

A	a higher rate, because it includes a credit risk premium for the tenants	✓ Correct	Correct: higher rate, because the commercial tenants add a credit risk premium.
B	the same rate, because both include inflation and inflation uncertainty premiums		Both include inflation and inflation uncertainty, but the commercial lease also has the credit premium, so the rates are not the same.
C	a lower rate, because commercial rents rise faster	✗ Your pick	Your pick: rents rising faster would raise the cash flows (the numerator), not lower the discount rate.

Trigger “a building leased to a government agency”
“a similar building leased to commercial tenants”

Rule next time: Same building, different tenant credit quality: the commercial lease carries an extra credit risk premium φ, so its rents are discounted at a HIGHER rate.

FORMULA / RULE

Discount rate = Rf + π + θ + φ (inflation, inflation uncertainty, other risks such as default)
Value = net rental income ÷ (1 + r)^t

WHY THE ANSWER IS A

1. Government tenant: very low default risk, small φ.
2. Commercial tenants: higher default risk, larger φ → higher r.
3. **A**.

Apply to other questions: Tenant credit risk → discount rate (φ). Growth in rents → cash flows.

Set 3 Q50 **Credit spread from CVA: LGD = 1 − recovery rate**

You **A** · Answer **B**

FI

VIGNETTE (RELEVANT PART, UNEDITED)

Marangoni next asks Brandt to analyze a three-year bond, issued by Norvalle S.A., using an arbitrage-free framework. The bond's coupon rate is 4%, with interest paid annually and a par value of 100. In the analysis, Brandt makes the following three assumptions:

The annual interest rate volatility is 15%.

The recovery rate is one-quarter of the exposure each period.

The annual probability of default each year is 1.50%.

Exhibit 2

Par Curve Rates for Annual Payment Benchmark Government Bonds

Maturity	Coupon rate	Price	Discount factor	Spot rate	Forward rate
1	2.50%	100	0.975610	2.5000%	2.5000%
2	3.30%	100	0.936888	3.3133%	4.1331%
3	4.00%	100	0.887981	4.0396%	5.5076%

Exhibit 3

One-Year Binomial Interest Rate Tree for 15% Volatility (risk-neutral probability of 50% on each branch)

Date	0	1 (top)	1 (bottom)	2 (top)	2 (middle)	2 (bottom)
Rate	2.5000%	4.7526%	3.5208%	7.2932%	5.4029%	4.0026%

Question 50. Based on Brandt's assumptions and Exhibits 2 and 3, the credit spread on the Norvalle bond is closest to:

OPTIONS

A	0.7796%.	✗ Your pick	Your pick: 0.7796% is what you get with a loss given default of 50%, not 75%.
B	1.1782%.	✓ Correct	Correct: LGD 75% gives a spread of 1.1782%.
C	1.5828%.		1.5828% assumes no recovery (LGD 100%).

Trigger “The recovery rate is one-quarter of the exposure each period.”
“The annual probability of default each year is 1.50%.”
“the relevant binomial interest rate tree is presented in Exhibit 3”: value the bond on the tree (VND)

Rule next time: Recovery of one-quarter means LGD = 1 − 1/4 = 75%. Then CVA = Σ DF × PD × LGD × expected exposure, fair value = VND − CVA, spread = YTM(fair value) − benchmark YTM.

FORMULA / RULE

LGD = 1 − recovery rate
CVA = Σ DF(benchmark) × PD(t) × LGD × expected exposure(t)
PD(t) = h × (1 − h)^(t−1) for a constant hazard rate h
Spread = YTM of (VND − CVA) − benchmark YTM

WHY THE ANSWER IS B

1. VND from the tree = 100.0000 (coupon equals the 3-year par rate).
2. CVA = 1.1250 + 1.0648 + 1.0080 = 3.1978.
3. Fair value = 96.8022, YTM = 5.1782%.
4. Spread = 5.1782% − 4.0000% = 1.1782%: **B**.

Apply to other questions: Recovery rate given → convert to LGD before anything else; read “one-quarter of the exposure” as recovery, not loss.

Mock 10 Q1 **Financial leverage = total assets ÷ equity, and the table's columns run from the newest year to the oldest**

You **C** · Answer **B**

FSA

VIGNETTE (RELEVANT PART, UNEDITED)

Greta Walsh is a financial analyst who is examining the most recent annual report of Westbrook Inc., a US-based company that complies with US GAAP. Walsh wants to determine Westbrook's ROE as well as the trend in its ROE components by using DuPont analysis. Data used for this purpose is presented in Exhibit 1.

Exhibit 1

	Most Recent Year	Previous Year	Two Years Ago
Revenue (USD million)	820	740	660
Total liabilities (USD million)	270	245	220
Total asset turnover ratio	1.40	1.35	1.30

Question 1. Based on the 3 years of data provided for DuPont analysis, Westbrook's financial leverage has been:

OPTIONS

A	stable.		Stable is wrong: the ratio moves in one direction (1.765 → 1.808 → 1.855).
B	trending higher.	✓ Correct	Correct: leverage is trending higher.
C	trending lower.	✗ Your pick	Your pick: leverage looks like it falls only if the columns are read left to right as if time ran forward. Here the left column is the MOST RECENT year.

Trigger "Most Recent Year | Previous Year | Two Years Ago" (the newest year is in the FIRST column) "trend in its ROE components by using DuPont analysis"

Rule next time: Leverage = assets ÷ equity, with assets = revenue ÷ asset turnover and equity = assets − liabilities. Then read the years in time order: the table runs from the most recent year back to two years ago, so a value that is higher in the first column has been rising.

FORMULA / RULE

> Total assets = revenue ÷ total asset turnover
> Equity = total assets − total liabilities
> Financial leverage = total assets ÷ equity

WHY THE ANSWER IS B

1. Assets: 820/1.40 = 585.7; 740/1.35 = 548.1; 660/1.30 = 507.7 (most recent, previous, two years ago).
2. Equity = assets − liabilities: 315.7; 303.1; 287.7.
3. Leverage: 1.855 (most recent), 1.808, 1.765 (two years ago): rising over time. **B**.

Apply to other questions: Before reading any trend, check the column order of the table: newest year first, or oldest year first.

Mock 10 Q9 **Standard I(B): paid travel from a covered company impairs independence; a small branded gift is acceptable**

You **A** · Answer **B**

Ethics

VIGNETTE (RELEVANT PART, UNEDITED)

Rodrick Darko, CFA, is a senior research analyst at TTN Capital, a large full-service brokerage and investment firm. Darko is invited to an analyst conference hosted by Bloom Data Centers, a company covered by Darko. Bloom arranges to pay for the commercial flights for all analysts attending the conference. Bloom also provides a company-branded mug as a gift to all analysts who attend the conference. Darko accepts the travel arrangement and the gift.

Question 9. Does Darko violate the Standards regarding the Bloom conference?

OPTIONS

A	No	✗ Your pick	Your pick: No. It does not help that the flights were offered to ALL analysts or that the decision would not change: I(B) asks whether the benefit could REASONABLY BE EXPECTED to compromise independence (appearance counts, not only actual influence), and the guidance says to travel at your own or your firm's expense instead of accepting paid travel from a company you cover.
B	Yes, by accepting the travel arrangement only	✓ Correct	Correct: a violation by accepting the travel only. The mug is a token gift.
C	Yes, by accepting the travel arrangement and the gift		Both is wrong: the mug alone does not compromise independence.

Trigger “Bloom arranges to pay for the commercial flights for all analysts attending the conference” “company-branded mug as a gift”

Rule next time: Do not accept travel paid by a company you cover: use commercial transportation at your own or your firm's expense. A token item such as a branded mug is a modest gift and is acceptable.

FORMULA / RULE

I(B) Independence and Objectivity: no gifts or benefits that could reasonably be expected to compromise independence

Best practice: pay for your own travel; small token gifts are fine

WHY THE ANSWER IS B

1. The company pays for the flights of analysts who cover it: this can appear to compromise Darko's independence.
2. The branded mug has little value: not a violation.
3. Violation by accepting the travel only: **B**.

Apply to other questions: Ask who pays and how much it is worth. A covered company paying for travel or lodging is a problem even when it is offered to everyone and even when you say it will not change your view; a small token gift is not. The exam follows the Standards' guidance, not the member's own judgment of whether he is influenced.

Mock 10 Q13 **Random walk written as an AR(1): intercept near 0 and slope near 1**

You **C** · Answer **A**

Quant

VIGNETTE (RELEVANT PART, UNEDITED)

Rafael Duarte, a research analyst at Lisbon Securities, is studying the time-series behavior of price-to-earnings ratios (P/Es) computed with trailing 12-month earnings (Etrailing). He and his assistant, Priya Nair, are reviewing the results of the ordinary least squares time series regression shown in Exhibit 1.

Exhibit 1

Results of Regression of P/E on Lagged P/E (P/Et = b0 + b1P/Et–1 + εt)

	Coefficient	Standard Error	t	Significance of t
Constant (b0)	0.118	0.162	0.728	0.234
Lagged P/E (b1)	0.994	0.003	331.333	0
R2	Standard Error of the Estimate	Durbin–Watson	F	Significance of F
0.082	1.51204	2.067	142.410	0

Duarte states: "This regression is a special case of a first-order autoregressive [AR(1)] model in which the value for b0 is close to zero and the value of b1 is close to 1. These values suggest that the time series is a random walk."

Question 13. Duarte's statement that the coefficients depicted in Exhibit 1 are consistent with a random walk is most likely:

OPTIONS

A	correct.	✓ Correct	Correct: the statement is right.
B	incorrect because b0 should be close to 1.		b0 close to 1 is wrong: the intercept of a random walk is 0.
C	incorrect because b1 should be close to 0.	✗ Your pick	Your pick: a slope close to 0 describes a series with almost no memory of its past value (no dependence on the lag), not a random walk.

Trigger "the value for b0 is close to zero and the value of b1 is close to 1"
Exhibit 1: b0 = 0.118 and b1 = 0.994
Rule next time: A random walk is x(t) = x(t−1) + ε. Estimated as an AR(1), that means b0 ≈ 0 and b1 ≈ 1, which is what Exhibit 1 shows.

FORMULA / RULE

Random walk: x(t) = x(t−1) + ε → b0 = 0, b1 = 1
With b1 = 1 the series has a unit root and no finite mean-reverting level b0 ÷ (1 − b1)

WHY THE ANSWER IS A

1. Exhibit 1 reports b0 = 0.118 (close to 0) and b1 = 0.994 (close to 1).
2. These values fit a random walk: **A**.

Apply to other questions: Slope near 1 = the next value is almost the last value (random walk); slope near 0 = no dependence on the past.

Mock 10 Q14 **Random walk: the best forecast of the next value is the current value**

You **A** · Answer **A** · right, not fully sure

Quant

VIGNETTE (RELEVANT PART, UNEDITED)

Nair replies: "I'm convinced the P/E series based on trailing earnings truly is a random walk."

Question 14. If Nair is correct regarding the trailing P/E time series, the best forecast of next period's trailing P/E is most likely to be the:

OPTIONS

A	current period's trailing P/E.	✓ Correct	Correct: the current period's value is the best forecast.
B	forecast derived from applying the AR(1) model depicted in Exhibit 1 to the data.		The AR(1) in Exhibit 1 gives almost the same number here (b0 ≈ 0, b1 ≈ 1), but a random walk is not covariance stationary, so AR(1) is not a valid model for it.
C	average P/E of the time series.		The average is the forecast for a mean-reverting series; a random walk has no mean-reverting level.

Trigger "I'm convinced the P/E series based on trailing earnings truly is a random walk"

Rule next time: If the series is a random walk, x(t+1) = x(t) + ε, so the best forecast is simply today's value. There is no mean to revert to and the series is not covariance stationary, so an AR(1) forecast or the series average is not appropriate.

FORMULA / RULE

> Random walk: x(t) = x(t−1) + ε → forecast = x(t)
> Mean-reverting level b0 ÷ (1 − b1) does not exist when b1 = 1

WHY THE ANSWER IS A

1. Nair says the series is a random walk.
2. For a random walk the best forecast is the last observed value: **A**.

Apply to other questions: "Random walk" → forecast = current value; do not use an average or an AR(1) forecast.

Mock 10 Q15 **ARCH(1) test: regress squared residuals on their own lag; the SLOPE c1 must differ from 0**

You **C** · Answer **C** · right, not fully sure

Quant

VIGNETTE (RELEVANT PART, UNEDITED)

Nair and Duarte next examine the behavior of P/Es calculated using forward 12-month earnings (Eforward). Nair estimates another AR(1) model but uses the forward P/E values this time. She denotes the errors from this second regression as ηt. She states: "The presence of first-order autoregressive conditional heteroskedasticity [ARCH(1)] errors in this regression is highly likely given the results reported in Exhibit 2."

Exhibit 2

Results of Regression of Squared Residuals, η_t^2, on Lagged Squared Residuals, η_{t-1}^2 ($\eta_t^2 = c0 + c1\eta_{t-1}^2 + ut$)

	Coefficient	Standard Error	t	Significance of t
Constant (c0)	0.312	0.041	7.610	0
Lag 1 (c1)	0.241	0.025	9.640	0
R2	Standard Error of the Estimate	Durbin–Watson	F	Significance of F
0.058	1.51204	2.067	92.930	0

Question 15. The results depicted in Exhibit 2 are best described as consistent with a regression that has ARCH(1) errors because:

OPTIONS

A	c1 is significantly different from 1.		c1 is tested against 0, not against 1.
B	c0 is significantly different from 0.		The intercept c0 does not test ARCH.
C	c1 is significantly different from 0.	✓ Correct	Correct: c1 is significantly different from 0 (t = 9.64).

Trigger "Regression of Squared Residuals, η_t^2, on Lagged Squared Residuals, η_{t-1}^2"
Lag 1 (c1): 0.241, t = 9.640, significant

Rule next time: ARCH(1) means today's error variance depends on yesterday's. Test it by regressing the squared residuals on the lagged squared residuals: if c1 is significantly different from 0, ARCH(1) errors are present.

FORMULA / RULE

ARCH(1) test: $\varepsilon(t)^2 = c0 + c1\,\varepsilon(t-1)^2 + u$
H0: c1 = 0 (no ARCH). Reject → ARCH present (then use ARCH/GARCH models or robust errors)

WHY THE ANSWER IS C

1. Exhibit 2 regresses the squared residuals on their own lag.
2. c1 = 0.241, t = 9.640 → significantly different from 0.
3. ARCH(1) errors are present: **C**.

Apply to other questions: Which coefficient decides? The SLOPE on the lag, and the null is 0.

Mock 10 Q23 **Option time decay: value one day later = value today + daily theta (theta is negative)**

You **B** · Answer **B** · right, not fully sure

Deriv

VIGNETTE (RELEVANT PART, UNEDITED)

Liu sells 12 call option contracts on a non-dividend-paying stock, Hillcrest Corporation, at the value determined by the BSM model, and considers delta-hedging the position using Hillcrest shares. Each call option contract is for 100 shares. Liu collects the information in Exhibit 1 for a call option on Hillcrest.

Exhibit 1

Current share price of Hillcrest	$62
Exercise price of Hillcrest call option	$60
d1	0.231
d2	0.082
N(d1)	0.591
N(d2)	0.533
Gamma	0.041
Theta (daily)	–0.021
BSM call value	$4.512

Question 23. Holding all else constant, the value of the call on Hillcrest one day after Liu's sale will be closest to:

OPTIONS

A	$4.471.		Uses gamma (0.041) instead of theta: 4.512 − 0.041 = 4.471.
B	$4.491.	✓ Correct	Correct: 4.512 − 0.021 = 4.491.
C	$4.533.		Adds the theta: 4.512 + 0.021 = 4.533, the wrong sign.

Trigger “Theta (daily) | –0.021”
“one day after Liu's sale”

Rule next time: Theta is the change in option value per day, all else equal. It is negative for a long call, so the value falls: 4.512 + (−0.021) = 4.491. Do not use gamma (that is for the delta change) and do not add the absolute value.

FORMULA / RULE

Value after 1 day ≈ value today + theta (daily)
Long call: theta negative (time decay); gamma = change of delta, not of value over time

WHY THE ANSWER IS B

1. Daily theta is −0.021.
2. New value = 4.512 − 0.021 = $4.491: **B**.

Apply to other questions: Time passes → use theta, with its sign. Price moves → use delta (and gamma).

Mock 10 Q24 **American put on a binomial tree: at every node take the larger of the continuation value and the exercise value**

You **A** · Answer **B**

Deriv

VIGNETTE (RELEVANT PART, UNEDITED)

Liu next considers an American-style put option on Marlow Corporation that matures in two years. She values it with a two-period binomial option pricing model and collects the relevant information in Exhibit 2. Marlow does not pay dividends.

Exhibit 2

Current share price of Marlow	$80
Exercise price of Marlow put option	$84
Risk-free rate (compounded annually)	4%
Up factor (u)	1.25
Down factor (d)	0.80
Stock prices in one year	S+ = $100, S- = $64
Stock prices in two years	S++ = $125, S+- = $80, S-- = $51.2

Question 24. Using the two-period binomial option pricing model, the value of the American put on Marlow Corporation is closest to:

OPTIONS

A	$8.45.	✗ Your pick	Your pick: $8.45 is the European put value. It never uses early exercise.
B	$9.89.	✓ Correct	Correct: $9.89, with early exercise at the down node.
C	$10.48.		$10.48 uses a 0.5 risk-neutral probability instead of 0.5333.

Trigger “American-style put option on Marlow Corporation that matures in two years”
“Exercise price of Marlow put option | $84”

Rule next time: For an American option compare, at each node, the discounted expected value with the exercise value K − S, and keep the larger. At the down node 84 − 64 = 20 beats the binomial value of 16.77.

FORMULA / RULE

q = (1 + r − d) ÷ (u − d) = (1.04 − 0.80) ÷ (1.25 − 0.80) = 0.5333

Node value = max[(q × up + (1 − q) × down) ÷ (1 + r), exercise value]

WHY THE ANSWER IS B

1. Puts at T = 2: Puu = 0 (84 − 125 < 0), Pud = 84 − 80 = 4, Pdd = 84 − 51.2 = 32.8.
2. Up node: (0.4667 × 4) ÷ 1.04 = 1.7949.
3. Down node: binomial value 16.7692, exercise 84 − 64 = 20, so P− = 20.
4. P0 = (0.5333 × 1.7949 + 0.4667 × 20) ÷ 1.04 = $9.89: **B**.

Apply to other questions: Word “American” → test early exercise at every node. A European answer is always a trap option.

Mock 10 Q37 **Unreimbursed expenses: compare them as a share of income, not in dollars**

You **A** · Answer **B**

Alt

VIGNETTE (RELEVANT PART, UNEDITED)

Exhibit 1

Selected Property Data

Property	Property 1	Property 2	Property 3
Property type	Downtown Office	Grocery-Anchored Retail Center	Multi-Family Building
Location	New York City	Miami	Boston
Age	10 years	7 years	5 years
Occupancy	88.00%	94.00%	96.00%
Gross potential rental income	$4,400,000	$1,950,000	$3,300,000
Loss to lease	($220,000)	($80,000)	($165,000)
Gross rental income	$4,180,000	$1,870,000	$3,135,000
Vacancy loss	($501,600)	($112,200)	($125,400)
Net rental income	$3,678,400	$1,757,800	$3,009,600
Expense reimbursement revenue	$290,000	$462,000	$0
Other income (includes % rent)	$410,000	$20,000	$52,000
Effective gross income	$4,378,400	$2,239,800	$3,061,600
Property management fees	($175,136)	($84,000)	($122,464)
Other operating expenses	($1,720,000)	($378,000)	($1,240,000)
Net operating income	$2,483,264	$1,777,800	$1,699,136

Martins asks Cho about the relative magnitudes of operating expenses in determining property NOI. Cho explains that the retail center in Miami is able to be fully reimbursed for its operating expenses but that the office property in New York City and the multi-family property in Boston must absorb considerable unreimbursed operating expenses.

Question 37. Which of the following statements is most accurate regarding the effect of unreimbursed operating expenses on the NOI of the New York City and Boston properties?

OPTIONS

A	The NOI of the office property in New York City is more negatively affected by unreimbursed operating expenses than is the NOI of the multi-family property in Boston.	✗ Your pick	Your pick: New York City has the larger expense in dollars, but it also has a much larger income base.
B	The NOI of the multi-family property in Boston is more negatively affected by unreimbursed operating expenses than is the NOI of the office property in New York City.	✓ Correct	Correct: Boston loses 44.5% of its effective gross income, New York City 36.7%.
C	The NOIs of both properties are affected similarly by unreimbursed operating expenses.		The two ratios are clearly different.

Trigger "must absorb considerable unreimbursed operating expenses"
rows: Expense reimbursement revenue, Effective gross income, Property management fees, Other operating expenses

Rule next time: Unreimbursed expenses = management fees + other operating expenses − expense reimbursement revenue. Compare that amount with effective gross income: the property with the larger share is hit harder.

FORMULA / RULE

> Unreimbursed expenses = management fees + other operating expenses − reimbursement revenue
> Effect on NOI = unreimbursed expenses ÷ effective gross income

WHY THE ANSWER IS B

1. New York City: (175,136 + 1,720,000 − 290,000) ÷ 4,378,400 = 36.7%.
2. Boston: (122,464 + 1,240,000 − 0) ÷ 3,061,600 = 44.5%.
3. Boston is hurt more: **B**.

Apply to other questions: Different-sized properties: always use a percentage, never compare raw dollar amounts.

Mock 10 Q38 **Real estate cycle: high vacancy means oversupply or recession, and recovery comes before expansion**

You **A** · Answer **C**

Alt

VIGNETTE (RELEVANT PART, UNEDITED)

Cho comments to Martins that these three properties may be in different phases of the real estate cycle. He specifically cites the differences between gross potential rental income and gross rental income in making his argument. In response to Cho's comments, Martins makes the two following statements about the office building in New York City:

Statement 1:

I believe that the high vacancy rates indicate that the net operating income of the office building has the potential to improve within two years.

Statement 2:

I believe that the office market in New York City is likely to enter the recovery stage of the real estate cycle within two years.

Question 38. Which of the following responses is most accurate regarding Martins's Statement 1 and Statement 2?

OPTIONS

A	Statement 1 is more accurate than Statement 2.	✗ Your pick	Your pick: Statement 1 expects NOI to improve already, which skips the recovery stage that must come first.
B	Statements 1 and 2 are both accurate statements.		Both cannot be accurate: they imply different phases of the cycle.
C	Statement 2 is more accurate than Statement 1.	✓ Correct	Correct: Statement 2 is more accurate.

Trigger Statement 1: "high vacancy rates indicate that the net operating income … has the potential to improve within two years"
Statement 2: "likely to enter the recovery stage of the real estate cycle within two years"

Rule next time: Cycle order: recovery → expansion → hyper-supply → recession. High vacancy puts the market in oversupply or recession, so the next stage is recovery, not expansion. Statement 2 names recovery; Statement 1 asks for improving NOI, which comes later (expansion).

FORMULA / RULE

> Cycle: recovery → expansion → hyper-supply → recession
> High vacancy = hyper-supply or recession; falling vacancy and rising rents = recovery and expansion

WHY THE ANSWER IS C

1. High vacancy points to recession or oversupply, the weak part of the cycle.
2. The next stage is recovery (Statement 2), where vacancy stops rising.
3. NOI growth comes mainly in the expansion that follows: Statement 1 jumps ahead.
4. Statement 2 is more accurate: **C**.

Apply to other questions: Place the market in the cycle first, then pick the next stage in order; do not skip a stage.

Mock 10 Q40 **Direct capitalization uses one period of income: use a normalized NOI for a property whose income is temporarily disrupted**

You **A** · Answer **A** · right, not fully sure

Alt

VIGNETTE (RELEVANT PART, UNEDITED)

Cho comments to Martins that these three properties may be in different phases of the real estate cycle. He specifically cites the differences between gross potential rental income and gross rental income in making his argument. In response to Cho's comments, Martins makes the two following statements about the office building in New York City:

Statement 2:

I believe that the office market in New York City is likely to enter the recovery stage of the real estate cycle within two years.

Martins asks Cho about the appropriateness of various valuation methods on these three properties. Cho suggests that the sales comparison approach is likely to be useful, but he is particularly interested in seeing the valuation results from an income approach for each of the three properties.

Question 40. Given Martins's comment in Statement 2 about the office property in New York City, which of the following approaches to estimating property value is most appropriate?

OPTIONS

A	Direct capitalization method using a normalized NOI forecast	✓ **Correct**	Correct: direct capitalization with a normalized NOI forecast.
B	Sales comparison approach		The sales comparison approach is the weaker choice while the market is still moving through the cycle (comparable sales may be few or reflect different conditions); Cho also wants an income approach.
C	Direct capitalization method using last year's NOI		Last year's NOI is not normalized when income is disrupted by the cycle.

Trigger Statement 2: "likely to enter the recovery stage of the real estate cycle within two years"
"he is particularly interested in seeing the valuation results from an income approach"

Rule next time: Direct capitalization = NOI ÷ cap rate, based on one year of income. If NOI is temporarily depressed by the cycle, last year's NOI misstates the value: use a normalized (stabilized) NOI, or a DCF.

FORMULA / RULE

Direct capitalization: value = NOI ÷ cap rate (single period)

Cycle disruption → normalized NOI or DCF

WHY THE ANSWER IS A

1. The office market is expected to recover in two years: today's NOI is temporarily depressed.
2. A one-period capitalization needs a normalized NOI: **A**.

Apply to other questions: Single-period method + temporarily abnormal income → normalize the income first.